

Jacob D. J. Peters

CONTACT INFORMATION

360 Prospect St
The Forest School at the Yale School of the Environment
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RESEARCH INTERESTS

My research focuses on understanding a wide range of forest phenomena at varying scales to inform forest management and climate smart forestry by combining in situ forest measurements and experiments with big data techniques and remote sensing. I currently study temperate liana (woody vine) ecology, impacts, management feasibility and benefits, and distributions using field experiments as well as satellite and UAV remote sensing. I am also involved in guiding several projects with masters of forestry and PhD students. These projects range from mapping and understanding biomass changes in forest fragments surrounding cocoa plantations in Ghana, understory regeneration in teak plantations in Ghana, Sal forest cover change in Bangladesh, the effect of fuel reduction treatments on wildfire severity in California, and street tree diversity in urban canopies of New England.

EDUCATION

Ph.D., Remote Sensing and Applied Forest Ecology, **May 2027**
Yale School of the Environment, New Haven, CT, USA

Dissertation Advisor: Profs. Mark Ashton and Xuhui Lee

Dissertation Title: *Temperate liana distributions, impacts, and remote sensing*

M.S., Forest Ecology, James Madison University, Harrisonburg, VA, USA **May 2020**

Honors: *Summa cum laude*, outstanding first year graduate student

Thesis Advisor: Dr. Heather Griscom

B.S., Environmental Science, UVA at Wise, Wise, VA, USA **May 2018**

Honors: *Magna cum laude*, Joseph C. Smiddy Outstanding Scientist Award

EMPLOYMENT

Teaching Fellow, Yale University, New Haven, CT, USA **January 2024–Present**

Drone Remote Sensing (ENV 704) Consultant, Yale University, New Haven, CT, USA **August 2024–December 2024**

Geospatial Consultant, Yale University, New Haven, CT, USA **August 2023–August 2025**

Adjunct Instructor, Ecology, James Madison University, Weyers Cave, VA, USA **January 2022–May 2022**

Adjunct Instructor, Biology, Blue Ridge Community College, Weyers Cave, VA, USA **August 2020–May 2022**

GIS Intern, Smithsonian Conservation Biology Institute, Front Royal, VA, USA **March 2021–December 2021**

Teaching Assistant, James Madison University, Harrisonburg, VA, USA **August 2018–May 2020**

Vegetation Survey Crew Lead, The Pennsylvania State University, State College, PA, USA **May 2018–July 2018**

Assistant Ecological Steward, UVA at Wise, Wise, VA, USA **November 2016–May 2018**

Park Ranger Career Path Intern, Virginia State Parks, USA **May 2015–August 2016**

HONORS, AWARDS, & SERVICE **CV Reviewer and Mock Interviewer**, Ecological Society of America Annual Meeting **2026**

Graduate Student Representative, Yale School of the Environment Faculty Search Committee **2025**

Student presentation winner (2nd), Society of American Foresters Natl. Convention **2025**

Best doctoral student presentation, Yale School of the Environment Research Day **2023**

Outstanding first-year graduate student, James Madison University **2019**

Three Minute Thesis People's Choice Award, James Madison University **2019**

Joseph C. Smiddy Outstanding Scientist Award, UVA at Wise **2018**

GRANTS & SCHOLARSHIPS **\$4,000** — Maryland Native Plant Society Research Grant **2026**

\$5,000 — Yale Center for Geospatial Solutions Affiliated Students Grant **2025**

\$20,000 — Childs Forestry Research Fund, The Forest School at the Yale School of the Environment **2024**

\$525,495 — Yale School of the Environment Doctoral Fellowship **2022–2027**

\$200,000 — Yale Center for Natural Carbon Capture, Yale School of the Environment **2022**

\$3,000 — James Madison University Biology Department Summer Support Award **2019**

\$1,000 — Norlyn L. Bodkin Botany Scholarship, James Madison University **2019**

\$48,000 — James Madison University Graduate Teaching Fellowship **2018–2020**

TEACHING EXPERIENCE **Teaching Fellow**, Yale School of the Environment

Science to Solutions: New Haven (Summer Orientation Module for incoming students) **Fall 2026**

ENV 659, The Practice of Silviculture: Principles in Applied Forest Ecology **Spring 2026**

ENV 660, Forest Dynamics **Fall 2025**

Yale Geographic Information Systems Accelerator **Spring 2024, 2025, 2026**

ENV 726, Observing Earth From Space **Spring 2024, 2025**

ENV 955, Seminar in Research Analysis & Communication in Forest Ecology **Spring 2024**

Guest Lectures

FORS 3100C, Mensuration and Biometrics, NMHU **Spring 2026**

ENV 751, Sampling Methodology and Practice, YSE **Spring 2026**

ENV 704, Workshop on Remote Sensing and Photogrammetry with Drones, YSE **Fall 2023, Fall 2024**

ENV 955, Seminar in Research Analysis & Communication in Forest Ecology, YSE **Spring 2024**

Adjunct Instructor, James Madison University

BIO 250L, Quantitative Ecology and Evolution Lab **Spring 2022**

Adjunct Instructor, Blue Ridge Community College, VA

BIO 101, General Biology I **Fall 2020–Spring 2022**

Teaching Assistant, James Madison University

BIO 140L, Foundations of Biology Lab **Fall 2018–Spring 2020**

MENTORING &
SUPERVISION

Graduate students

Akshay Patel, Yale School of the Environment **Fall 2025–Present**

Sierra Chadwick, Yale School of the Environment **Fall 2025–Present**

Louis Cimmino, Yale School of the Environment **Fall 2025–Present**

Kumba Jammeh, Yale School of the Environment **Fall 2024–Present**

Nate McMullen, Yale School of the Environment **Fall 2024–2025**

Jackson Cooper, Yale School of the Environment **Summer 2024–2025**

Cindy Cifuentes, Yale School of the Environment **Summer 2024–2025**

Yeim We, Yale School of the Environment **Summer 2024**

Sangam Paudel, Yale School of the Environment **Summer 2024**

Michael Freiburger, Yale School of the Environment **Fall 2022–Spring 2023**

Undergraduate students

Riley Johnson, Washington College, MD **Summer 2024**

Emilie Fowler, James Madison University, VA **Fall 2018–2019**

Zachary Warning, SUNY ESF, NY **Summer 2018**

Kaj Overturf, University of Maine, ME **Summer 2018**

Rosa Kome, Clemson University, SC **Summer 2018**

MEMBERSHIPS

Member, Ecological Society of America **2026–Present**

Member, Society of American Foresters **2025–Present**

Member, Yale Graduate Student Forestry Club **2023–Present**

Member, Yale Center for Geospatial Solutions Affiliated Students **2024–Present**

President, UVa-Wise Environmental Club **Spring 2017–2018**

Darden Society member **2016–Present**

COMPUTER SKILLS **Languages**—Expert in R. Proficient in Javascript, Bash, Python, Git.
Markup languages: L^AT_EX, HTML, Markdown.

Software—ArcGIS, QGIS, ENVI, Google Earth Engine, Agisoft Metashape, OpenDroneMap, DJI Terra, CloudCompare, R, Adobe Photoshop and Illustrator, GIMP, Inkscape, GitHub, Microsoft Suite.

Operating systems— Windows, Mac OS, Linux.

**MANUSCRIPTS IN
PROGRESS**

8. **Peters, J. D. J.**, Louis Cimmino, Nate McMullen, Manuel Flores, Michael Remke, David Woodbury (2026) *Forest fuels management effect on wildfire severity in the United States*. In Preparation.
7. **Peters, J. D. J.**, Xuhui Lee, Bronson W. Griscom, Grant Connette, Mark S. Ashton (2026) *Quantifying Liana Infestation Through UAV Multispectral and LiDAR Surveys*. In Preparation.
6. **Peters, J. D. J.**, Xuhui Lee, Mark S. Ashton (2026) *Spatial Taxonomy and Phylogeny of the World's Lianas*. In Preparation.
5. Nyanzu, B. A., Kumba Jammeh, David Woodbury, **Jacob D. J. Peters**, Hugh C. A. Brown, and Mark S. Ashton (2026) *Assessing the composition and structural degradation of tropical forest fragments within a cocoa-based agricultural landscape*. In Revision.
4. Jammeh, K., Mark S. Ashton, **Jacob D. J. Peters** (2026) *Assessment of Tree Growth Responses Following Liana Removal in Urban Woodlands*. In Preparation.
3. Flores, M., David Woodbury, **Jacob D. J. Peters**, Mark S. Ashton (2026) *Identifying potential tree species for reforestation from 40 years of growth measurements in a Sri Lankan mixed dipterocarp rainforest*. In Preparation.
2. Jammeh, K., Bernard A. Nyanzu, Mark S. Ashton, **Jacob D. J. Peters**, and Hugh C.A. Brown (2026) *Assessing the understory vegetation of teak (*Tectona grandis*) in relation to plantation age and ownership Ghana*. In Preparation.
1. **Peters, J. D. J.**, Xuhui Lee, Bronson W. Griscom, Grant Connette, Mark S. Ashton (2027) *Monitoring Tree Growth Responses to Liana Removal in Mid Atlantic Hardwoods Using Drone and Field Measurements*. In Preparation.

PUBLICATIONS

7. **Peters, J. D. J.**, Grant Connette, Xuhui Lee, Bronson W. Griscom, Mark S. Ashton (2026) *Eastern North American Liana Distributions Highlight Significant Opportunities for Improved Carbon Sequestration*, [Diversity and Distributions 32\(4\)](#). [Diversity and Distributions 32\(4\)](#).
6. Freiburger, M., Danica Doroski, Colleen M. Dunning, Mark S. Ashton, **Jacob D. J. Peters** (2025) *The 10/20/30 Planting Rule Aligns with Traditional Plant Diversity Metrics across Spatial Scales*, [Urban Forestry and Urban Greening 114](#).
5. Schnitzer, Stefan A., Begüm Kaçamak, Isaac Zombo, et al., including **Jacob D. J. Peters** ... (2025) *The Search for Champion Lianas: The Largest Lianas on Six Continents.*, [Biotropica 57\(6\)](#).
4. **Peters J.D.J.**, J.M. Portmann, B.W. Griscom (2023) *Lianas (*Vitis* spp.) reduce growth and carbon sequestration of light-demanding tree species in a temperate forest*, [Restoration Ecology 31\(5\)](#), 1-10.
3. **Peters J.D.J.**, H. Griscom (2022) *Highlighting an overlooked hotspot for American ginseng (*Panax quinquefolius* L.) in Virginia*, [Natural Areas Journal 42\(2\)](#), 209-119.
2. **Peters J.D.J.**, S. Schoen, M. Rhodes, H. Griscom (2021) *Up Smith Creek without a paddle: a case study on the barriers to stream restoration assessment*, [Journal of Ecological Restoration 39\(3\)](#), 151-155. [Alternative link to avoid paypal](#).

1. **Peters J.D.J.**, R. Huish, D. Taylor, B. Munson (2020) *Comparative Analysis of Four Maple Species for Syrup Production in South-Central Appalachia*, *Journal of Agriculture, Food Systems, and Community Development* 9(2), 267-276.

INVITED TALKS

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| 9. Yale Center for Geospatial Solutions' Website Building Workshop | December 2025 |
| 8. Yale Environmental Training Initiative's Forest Monitoring Workshop | June 2025 |
| 7. Smithsonian Conservation Biology Institute's Conservation Ecology Center | January 2025 |
| 6. "Upgoer 5" at Yale School of the Environment Research Day | April 2023 |
| 5. Smithsonian Conservation Biology Institute's Conservation Ecology Center | August 2021 |
| 4. Research Day, University of Virginia's College at Wise | April 2018 |
| 3. Virginia Master Naturalists, Clinch Valley Chapter | December 2017 |
| 2. Clinch River Valley Initiative Youth Summit | March 2017 |
| 1. Letcher Co. KY Maple School | December 2016 |

CONTRIBUTED
TALKS (SELECTED)

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| 15. Ecological Society of America Annual Meeting | July 2026 |
| 14. Ecological Society of America Annual Meeting | July 2026 |
| 13. Society of American Foresters National Convention | October 2025 |
| 12. New England Society of American Foresters Annual Meeting | March 2025 |
| 11. Yale Center for Geospatial Solution Affiliated Students Luncheon | October 2024 |
| 10. Yale Center for Natural Carbon Capture Showcase | September 2024 |
| 9. Yale Center for Natural Carbon Capture Showcase | October 2023 |
| 8. Yale School of the Environment Research Day | April 2023 |
| 7. Annual Conference for the Assoc. of Southeastern Biologists | April 2019 |
| 6. Annual Conference for the Assoc. of Southeastern Biologists | March 2018 |
| 5. Annual North American Agroforestry Conference | April 2017 |
| 4. National Conference on Undergraduate Research | April 2017 |
| 3. Annual Conference for the Assoc. of Southeastern Biologists | March 2017 |
| 2. Annual Conference Society of Economic Botany | June 2016 |
| 1. National Conference on Undergraduate Research | April 2016 |

REFERENCES

Dr. Mark S. Ashton, Professor of Silviculture and Director of Yale Forests, Yale School of the Environment
 email: mark.ashton@yale.edu
 office phone: 1-203-432-9835

Dr. Xuhui Lee, Professor of Meteorology, Yale School of the Environment
 email: xuhui.lee@yale.edu
 office phone: 1-203-432-6271

Dr. David Woodbury, Assistant Professor of Forestry, New Mexico Highlands University
 email: dwoodbury@nmhu.edu
 office phone: 1-505-426-2267